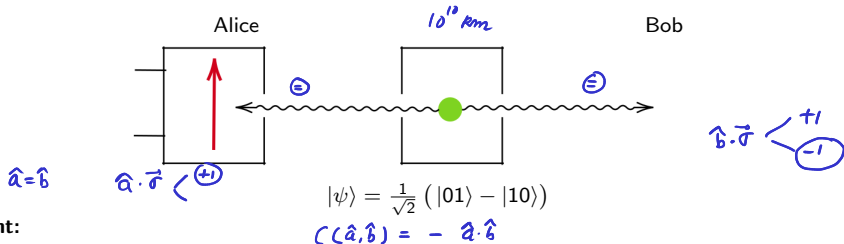


Q π _{AI} Bell inequality: the question is about the origin of probabilistic nature

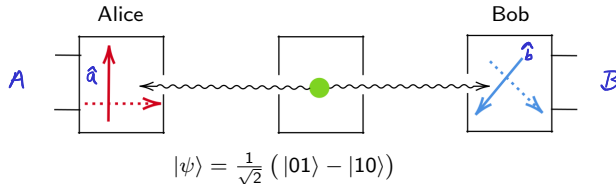


EPR Argument:

- The **result** of a measurement on Bob's particle must be unaffected by **operations** on Alice's particle that is **far far away** (spatially separated). (**Locality**) *meas-setting*
- Since we can **predict in advance** the result of measuring **any chosen** spin-component $\hat{b} \cdot \vec{\sigma}$ of Bob's particle (**element of reality**), by previously measuring the same component $\hat{a} \cdot \vec{\sigma}$, ($\hat{a} = \hat{b}$), of Alice's particle, it implies that the result of any such measurement must actually be **predetermined**. *Pre exist*
- Since the **initial** quantum state-vector $|\psi\rangle$ does not determine the result of an **individual** measurement, this **predetermination** suggests the possibility of a **more complete** description of the state.

Underlying thought: the probabilistic nature maybe comes from a lack of knowledge?

Q π_{AI} Bell inequality: the question is about the origin of probabilistic nature



- The **result** of a measurement on Bob's particle must be unaffected by **operations** on Alice's particle that is **far far away** (spatially separated). (**Locality**)
- Since the **initial** quantum state-vector $|\psi\rangle$ does not determine the result of an **individual** measurement, this **predetermination** implies the possibility of a **more complete** description of the state. (**Let us imagine a hidden variable λ whose value determines the outcomes on both sides. So each pair comes with a λ .**)

Probability distribution of λ for the statistical predictions: $p(\lambda) \geq 0$, $\int p(\lambda) d\lambda = 1$

Outcomes: $A(\hat{a}, \lambda) \in \{+1, -1\} \ni B(\hat{b}, \lambda)$

Locality $\swarrow$ $\nwarrow$ *determinism*

Expectation value: $\mathcal{C}(\hat{a}, \hat{b}) := \int A(\hat{a}, \lambda) B(\hat{b}, \lambda) p(\lambda) d\lambda$ (**NOT the Born rule**)

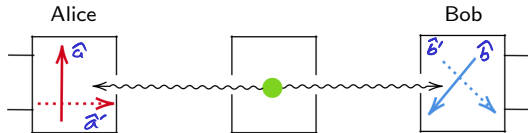
J. S. Bell, Physics **1**, 195 (1964).

B.-G. Englert, *Lectures on Quantum Mechanics - Vol 1: Basic Matters* (World Scientific, 2006), Chapt 1.

$Q\pi_{AI}$ Bell inequality: the question is about the origin of probabilistic nature

$$\left. \begin{array}{l} \hat{a} \cdot \vec{\sigma} \\ \hat{a}' \cdot \vec{\sigma} \end{array} \right\}$$

$$\left. \begin{array}{l} \hat{b} \cdot \vec{\sigma} \\ \hat{b}' \cdot \vec{\sigma} \end{array} \right\}$$



$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle - |10\rangle)$$

Same diff

$$A + A' \in \{\pm 2, 0\}$$

$$A - A' \in \{0, \pm 2\}$$

Outcomes: $A(a, \lambda) \in \{+1, -1\} \ni B(b, \lambda)$

Correlation: $\mathcal{C}(a, b) := \int A(a, \lambda) B(b, \lambda) p(\lambda) d\lambda$ (NOT the Born rule)

$$\left[A(a, \lambda) + A(a', \lambda) \right] B(b, \lambda) + \left[A(a, \lambda) - A(a', \lambda) \right] B(b', \lambda) = \pm 2 \quad (\text{values assigned to all the four observables simultaneously})$$

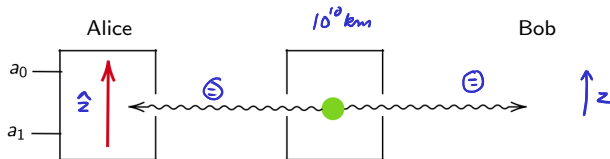
$$(A + A') B + (A - A') B' = \pm 2$$

$$\int AB + \int A'B + \int AB' - \int A'B'$$

CHSH Inequality: $\mathcal{C}(a, b) + \mathcal{C}(a', b) + \mathcal{C}(a, b') - \mathcal{C}(a', b') \leq 2$

J. F. Clauser, M. A. Horne, A. Shimony, and R. A. Holt, Phys. Rev. Lett. **23**, 880 (1969).
 B.-G. Englert, *Lectures on Quantum Mechanics - Vol 1: Basic Matters* (World Scientific, 2006), Chapt 1.
 A. Peres, *Quantum Theory: Concepts and Methods* (Kluwer Academic Publisher, 1995), Chapt 6.

NO faster than light communication



$$|\psi\rangle_{ab} = \frac{1}{\sqrt{2}} (|01\rangle_{ab} - |10\rangle_{ab})$$

$$|0\rangle_a |1\rangle_b$$

Alice's z-measurement outcome: 0

Then, Alice's **knowledge** of Bob's state: $\rho_b^{(alice)} = |1\rangle\langle 1| \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

Bob's **knowledge** of Bob's state: $\rho_b^{(bob)} = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|) \equiv \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$

After a call from Alice, Bob's **knowledge** of Bob's state: $\rho_b^{(bob)} = |1\rangle\langle 1| \equiv \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

Quantum state is not an objective property such as charge or mass of a particle, it represent our knowledge about the "preparation" of the system.

Schmidt decomposition of a bipartite state-vector

$$\{|i\rangle_a\} \otimes \{|j\rangle_b\}$$

$$\alpha_{ij} = \langle ij | \psi \rangle \in \mathbb{C}$$

$$\begin{pmatrix} 0 & 0 & 0 & 0 \\ & a & & \\ & & b & \\ & & & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 & 0 \\ & 0 & 0 & 0 \\ & & 0 & 0 \\ & & & 0 \end{pmatrix}$$

$$|\psi\rangle_{ab} = \sum_{i,j} \alpha_{ij} |i\rangle |j\rangle$$

$$= \sum_i |i\rangle_a \left(\sum_j \alpha_{ij} |j\rangle_b \right)$$

$$= \left(|1\rangle_a \cdots |d\rangle_a \right) \underbrace{\begin{pmatrix} \alpha_{11} & \cdots & \alpha_{1d} \\ \vdots & & \vdots \\ \alpha_{d1} & \cdots & \alpha_{dd} \end{pmatrix}}_{A = UDV} \begin{pmatrix} |1\rangle_b \\ \vdots \\ |d\rangle_b \end{pmatrix}$$

- Schmidt coefficients (singular values) are square-root of eigenvalues of $AA^\dagger$:

$$\zeta_1 \geq \zeta_2 \geq \cdots \geq 0$$

- If number of **non-zero** Schmidt coefficients is **more than one**, then $|\psi\rangle$ represents an entangled state.

- Putting Rest = 0 gives **quantum data compression**, that is applied to build tensor networks (for quantum machine learning).

- $|\psi\rangle$ is a **purification** of the mixed state ρ_{red} .

$$S(\rho_{\text{red}}) = - \sum_k \zeta_k^2 \ln(\zeta_k^2) > 0$$

$$\{|\phi\rangle_a\} \otimes \{|\chi\rangle_b\}$$

SVD

$$A = UDV$$

$$|\psi\rangle_{ab} = \sum_k \zeta_k |\phi_k\rangle |\chi_k\rangle$$

$$= \underbrace{\zeta_1 |\phi_1\rangle |\chi_1\rangle + \zeta_2 |\phi_2\rangle |\chi_2\rangle}_{|\psi_{ab}\rangle} + \underbrace{\text{Rest}}_0$$

$$\rho_{ab} = |\psi\rangle\langle\psi| = \sum_j \zeta_j^2 |\phi_j\rangle\langle\phi_j| \otimes |\chi_j\rangle\langle\chi_j| + \dots$$

$$\rho_{\text{red}} = \text{tr}_{\text{sub}}(|\psi\rangle\langle\psi|) = \sum_k \zeta_k^2 |\phi_k\rangle\langle\phi_k|$$

$$|\chi\rangle_b = \sum_j u_{ij} |j\rangle_b$$

$$|\phi\rangle_a = \sum_i u_{ij} |i\rangle_a$$

